

Project Design Phase

EduGenie – AI-Powered Learning Assistant

Problem-Solution Fit

Problem	EduGenie Solution
Difficult Concepts	Concept Explanation and Question & Answer
Self-Testing	Quiz Generator (3-question multiple-choice quiz)
Long Materials	Text Summarizer (bullet-point summaries)
Unstructured Learning	Personalized Learning Path (based on topic and level)

One AI-driven assistant addresses all four learning challenges through a single interface.

Proposed Solution

EduGenie is an AI-powered educational assistant powered by Google Gemini. It combines five essential study tools in one platform and is designed to make learning simpler, faster and more structured.

Five Core Study Tools

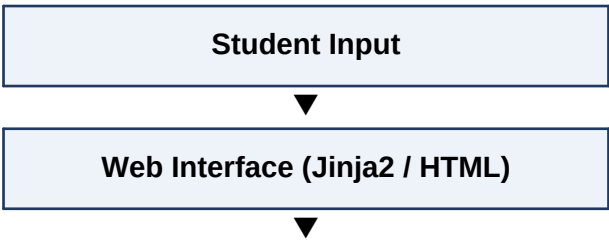
- 1 Question & Answer
- 2 Concept Explanation
- 3 Quiz Generator
- 4 Text Summarizer
- 5 Personalized Learning Path

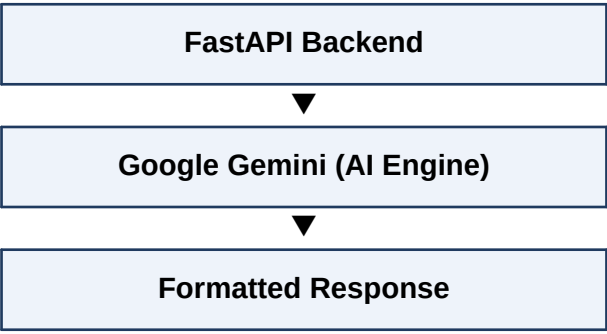
Project Goal

Give students one assistant for understanding, revision, self-testing and planning what to learn next.

Solution Architecture

System Workflow





Technology Stack

Component	Technology
Backend	FastAPI (Python)
AI Engine	Google Gemini
Frontend Templates	Jinja2 (HTML)
Server	Uvicorn